

12th International Workshop on Human Activity Sensing Corpus and Applications (HASCA)

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ABSTRACT

The recognition of complex and subtle human behaviors from wearable sensors will enable next-generation human-oriented computing in scenarios of high societal value (e.g., dementia care). This will require a large-scale human activity corpus and much-improved methods to recognize activities and the context in which they occur. This workshop deals with the challenges of designing reproducible experimental setups, running large-scale dataset collection campaigns, designing activity and context recognition methods that are robust and adaptive, and evaluating systems in the real world. We wish to reflect on future methods, such as lifelong learning approaches that allow *open-ended* activity recognition. This year HASCA will welcome papers from participants to the *Fifth Sussex-Huawei Locomotion and Transportation Recognition Challenge* and the *First WEAR Dataset Challenge* in special sessions.

CCS CONCEPTS

• General and reference → Surveys and overviews.

KEYWORDS

large-scale human activity sensing corpus, activity recognition, wearable computing, open-ended activity/context recognition, smartphones, mobile sensors, participatory sensing

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1 OBJECTIVE OF THE WORKSHOP

The objective of the workshop is to bring together researchers and practitioners both from academia and industries to discuss, identify, and share experiences surrounding the construction of human activity sensing corpora and their applications. Similarly to other human-related information processing areas, such as speech recognition and image recognition, real-world activity recognition requires large-scale corpora of data. Some initiatives set reference datasets for the use by the community, such as OPPORTUNITY [6], HASC [4], or SHL [2], but more work on larger scale and richer datasets are required. Scenarios of high societal value — such as healthcare — are likely to require a more complete and subtle understanding of the user’s activities, their quality, and the context in which they occur, beyond what is currently available in “off-the-shelf” datasets. Similarly, activity recognition methods will have to be further improved to tackle real-world challenges.

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After its success in previous years, the workshop will have a special session dedicated to the *Sixth Sussex-Huawei Locomotion and Transportation Activity Recognition Challenge* [5]. The competition is organized by the HASCA organizers using the *University of Sussex-Huawei Locomotion (SHL) dataset* [2, 5, 7], which is a large-scale dataset for multimodal locomotion analytics collected during 2017. In addition to the SHL Challenge, HASCA will introduce the *First WEAR Dataset Challenge*, a sports activity recognition challenge based on the WEAR dataset [1]. Participants of the challenge will be tasked to predict workout activities present in a yet unreleased test dataset of newly and re-recorded participants. Both the SHL and WEAR challenge will run a few months in advance of the workshop. Participants are encouraged to submit a paper to the HASCA workshop to present their results, where we will unveil which are the best-performing methods.

In addition, we are looking to evolve the workshop style toward increasing collaborations within the community on larger-scale tasks and challenges. Many domains have seen fruitful improvements with such community-wide collaborations, e.g., collecting joint datasets across researchers all over the world, using the same sensors, targeting the same tasks, and harmonizing these results. One example is the CVPR challenge. At the end of the workshop, the participants will discuss the next year's collaborative work based on the entered projects and aim for significant improvements of HASCA for the next year.

2 TOPICS OF THE WORKSHOP

The objective of this workshop is to share the experiences among current researchers around the challenges of real-world activity recognition, the role of datasets and tools, and breakthrough approaches toward open-ended contextual intelligence. We expect the following domains to be relevant contributions to this workshop (but not limited to):

- (1) *Data collection / Corpus construction methodologies*: Experiences or reports from data collection and/or corpus construction projects, such as formats or methodologies for data collection including participatory sensing and crowd-sourcing data collection. A wide abundance of sensor data is potentially in reach with users instrumented with their mobile phones and other wearables. Capitalizing on crowd-sourcing to create larger datasets cost-effectively may be critical to open-ended activity recognition.
- (2) *Effectiveness of Data / Data Centric Research*: Data-centric practices focus on creating the best dataset for a model in contrast to improving the model for a given dataset. Experiences in data-centric AI for HAR as well as using large-scale human activity sensing corpus can help inform how to evaluate a dataset and how to plan new experiments to obtain maximum benefits.
- (3) *Tools and Algorithms for Activity Recognition*: Activity recognition researchers could be more focused on their research theme with good tools for the management of sensor data. In this workshop, we solicit development reports of tools and algorithms for forwarding to the community.
- (4) *Real World Application and Experiences*: Activity recognition "in the Lab" usually works well. However, this is not true in

the real world. There is a huge gap/valley between the "Lab Environment" and the "Real World Environment". Reports on how large-scale human activity sensing corpus can help to overcome this gap/valley are welcome.

- (5) *Sensing Devices and Systems*: Data collection extends beyond the use of off-the-shelf sensors, necessitating the development of specialized devices to acquire specific types of information. We welcome papers focusing on specialized sensor development as well as developing or evaluating systems and technologies for data collection.
- (6) *Unsupervised pattern discovery*: Discovering meaningful repeating patterns in sensor data can be fundamental in informing other elements of a system generating an activity corpus, such as inquiring users or triggering annotation crowd-sourcing.
- (7) *Data augmentation, generation or simulation*: Collecting data and labeling is time-consuming and error-prone. Data augmentation, generation, or simulation techniques via generative AI, physics simulation, or modality transfer could potentially augment current datasets to cover a wide range of individual variations.
- (8) *Transfer learning, semi-supervised learning, incremental learning, lifelong learning*: The ability to translate recognition models across tasks or modalities or to use minimal supervision can enable the reuse of datasets across domains and reduce the costs of acquiring annotations.

3 WORKSHOP FORMAT

We plan for a 1-day workshop format and hold it on October 5th, 2024 in Melbourne, Australia. There will be a keynote talk by a practitioner. After this, there will be two technical paper sessions to present and discuss the accepted papers. Finally, we will have a session to present the challenge participation and results. There will be a discussion session after all of the presentations. In short, we plan up to three technical paper sessions and keynote talks in the one-day workshop.

4 ESTIMATED NUMBER OF PARTICIPANTS

We expect to bring together 40–50 participants who are working on or have an interest in human activity sensing corpus and its applications. The accepted paper count for HASCA2022 was 12, which is one of the biggest workshops in UbiComp/ISWC2022. Our workshop is open to anyone and we do not cap the number of participants because it is very important to share the experience and the information of the corpus in this field for real-world practical applications.

5 PAPER SELECTION AND PUBLICATION

All submissions will go through a light review process with one or two reviewers. All papers will be collected through Precision Conference system (PCS). All contributions will be included in the ACM Digital Library and supplemental proceedings of the conference. Our website will be <http://hasca2024.hasc.jp/>.

6 PROSPECTIVE OUTCOME

The workshop will provide knowledge about current state-of-the-art studies around human activity recognition and challenges in dataset creation with an open-ended concept for UbiComp/ISWC community. Expected outcomes are:

- (1) Survey of the state of the art of “Human Activity Sensing Corpus”. This includes an overview of new data collection methods, tools, and algorithms.
- (2) Practical knowledge of the data collection methodologies for human activity sensing.
- (3) Recognition of the potential and the importance of the large-scale corpus for human activity recognition.

7 PRE-WORKSHOP ACTIVITIES

We have already held the past ten HASCA workshops from 2013 through 2023. Every time, there are new attendees and the community has been growing continuously. We have created a mailing group¹ for the community. In 2019, a book summarizing the first five workshop outcomes was published [3].

8 POST-WORKSHOP ACTIVITIES

On the workshop website, the list of the current data collection activities and the information on the public corpora will be presented.

9 ORGANIZERS

- Kazuya Murao is an associate professor at Ritsumeikan University, Japan. He is working on wearable computing and human activity recognition and is a member of HASCA.
- Yu Enokibori is a lecturer at Nagoya University, Japan. He is working on wearable computing, elderly healthcare, and human activity recognition, and also is a member of HASCA.
- Hristijan Gjoreski is associate professor at the Ss. Cyril and Methodius University in Skopje, Macedonia. His expertise include wearable computing, machine learning, artificial intelligence, with focus on behaviour and emotion analytics.
- Paula Lago is an assistant professor at Concordia University in Canada. Her current research is on how to improve the generalization of activity recognition in real-life settings taking advantage of data collected in controlled settings.
- Tsuyoshi Okita is an associate professor at Kyushu Institute of Technology, Japan. He is interested in deep learning and its application to various areas, including activity recognition (signal and video).
- Pekka Siirtola is a post-doc researcher at University of Oulu, Finland. He is studying how to extract useful information from wellness data. Especially he is interested in continuous learning and wearable sensors.
- Kei Hiroi is an associate professor at Kyoto University, Japan. She is working on flood analysis/prediction and spatial-temporal analysis. She is a member of HASCA.
- Philipp M. Scholl currently works at the Embedded Systems, University of Freiburg, Germany. His interests are wearable computing and activity recognition.

- Mathias Ciliberto is a Ph.D. student at University of Sussex, UK. He is studying data mining and artificial neural network.
- Kenta Urano is an assistant professor in Graduate School of Engineering, Nagoya University. His research interests include location based system, human activity recognition, and biosignal entertainment computing.
- Marius Bock is a Ph.D. student at the University of Siegen, Germany. His research interests lie in wearable- and video-based human activity recognition.

10 CALL FOR PAPERS

HASCA 2024 is the eleventh workshop in this series, with a growing community of practitioners and researchers interested in the applications of human activity recognition. The goal of this workshop is to continue conversations regarding current challenges of real-world activity recognition with newly developed datasets and tools, breaking through towards open-ended contextual intelligence. This workshop discusses the challenges of designing reproducible experimental setups, running large-scale dataset collection campaigns, designing activity and context recognition methods that are robust and adaptive, and evaluating systems in the real world. This year, we wish to reflect on future methods, such as lifelong learning approaches that allow open-ended activity recognition. Papers may include topics such as:

- New Human Activity Sensing Corpus
- Large Scale, Longitudinal and Real-world Data Collection experiences
- Data Validation, Tagging / Labeling and Fusion
- Efficient Data Collection
- Data Mining from Corpus
- Automatic Segmentation
- Performance Evaluation
- Noise Robustness
- Non-Supervised Machine Learning for HAR
- Tools for Human Activity Corpus/Sensing/Analysis
- Participatory Sensing and Crowd-sensing for Data Collection
- Context Awareness
- Compressive Sensing
- Sensing Devices
- Lifelog Systems
- Wearable Application
- Deep Learning for HAR
- Human activity recognition applications such as Social Activities Analysis/Detection, Pedestrian Navigation, Gait Analysis, Daily-life Worker Support, Health-care Monitoring and Recommendation

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¹<https://groups.google.com/forum/#!forum/hasca-workshop>

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